Supporting Information

Structure-Aware Row Scaling for Block-Structured Mixed-Integer Programs

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This document contains (i) the 0.1%-MIPGap Gurobi distributional, diagnostic, and performance-profile views (Scenario 2 of the main text; Figures S1–S3), which mirror the corresponding 1%-MIPGap figures of the main text; (ii) the 0.1%-MIPGap CPLEX distributional and profile views (Scenario 4; Figure S4); (iii) the hyperparameter sensitivity of the structure-aware rule (Table S1); (iv) the per-variant original-scale reliability breakdown with PAR10 under the three readings of the main text (Table S2); (v) the out-of-sample sensitivity of the collective coupling reliability floor (Table S3); and (vi) the evaluation of the optional local reliability floor on the failing Simple subset (Table S4). The quantitative evidence for every scenario is in the main-text tables.

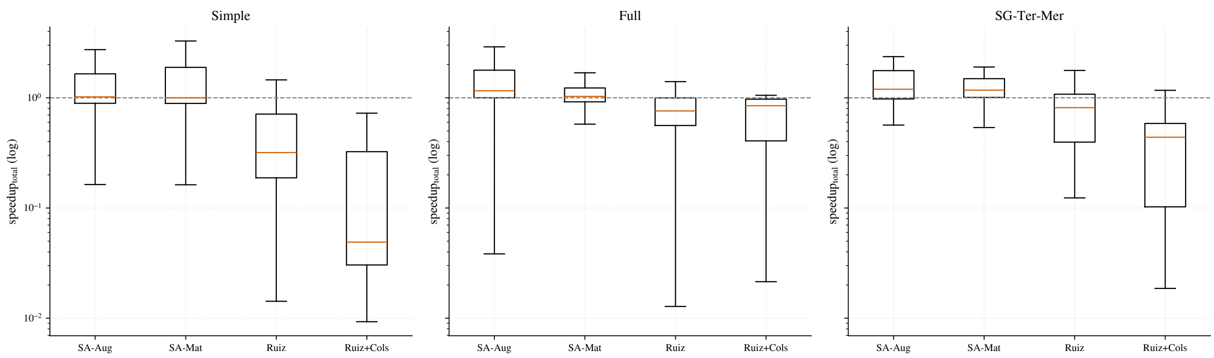
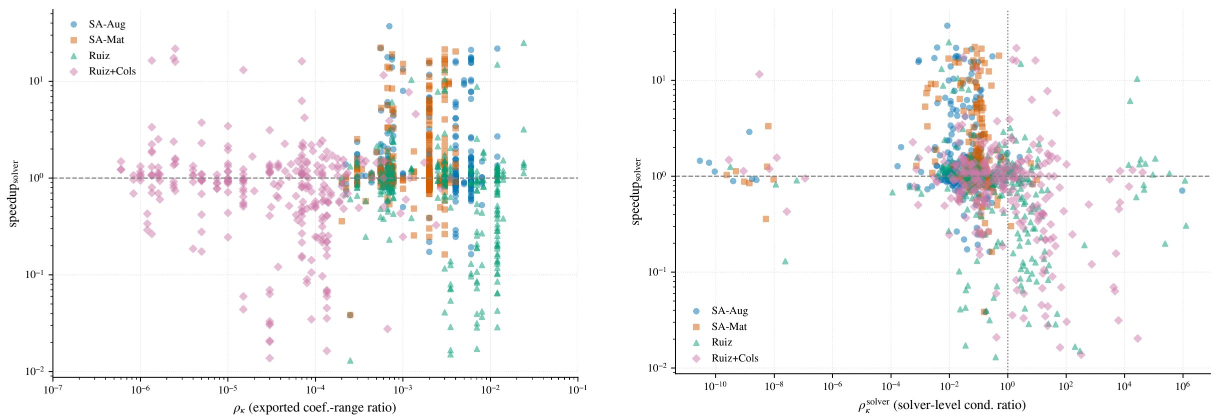


Figure S1: At 0.1% MIPGap, the Gurobi speedup distributions preserve the same qualitative pattern observed at the looser tolerance. Values above one indicate faster runs than the unscaled formulation.



(a) Static coefficient-range ratio.

(b) Solver-level conditioning ratio.

Figure S2: At 0.1% MIPGap, the stricter Gurobi runs again separate static coefficient compression from solver-facing numerical behavior. The solver-level diagnostic remains more informative than the exported coefficient-range ratio.

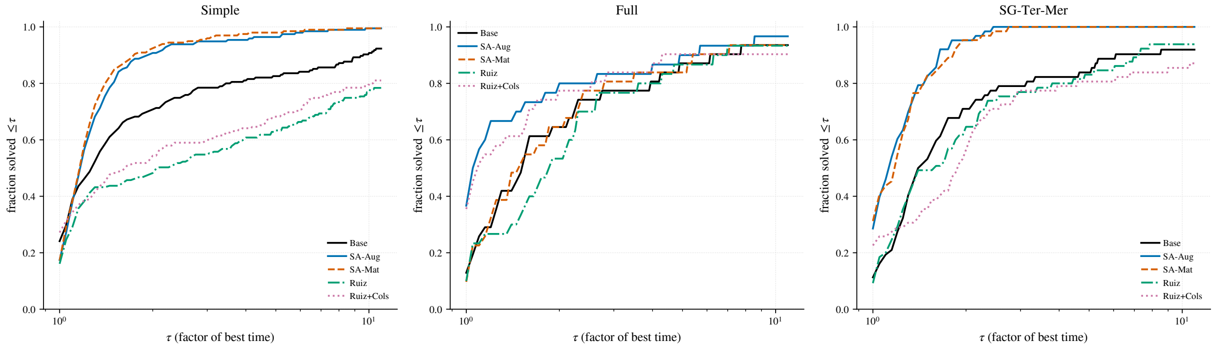
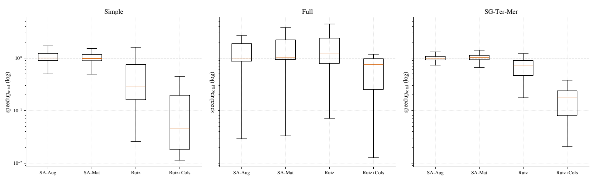
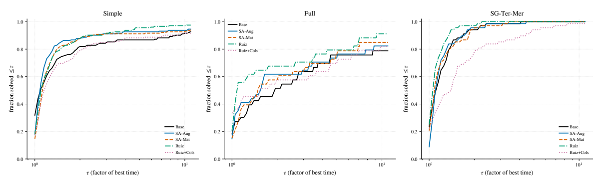


Figure S3: At 0.1% MIPGap, the Dolan–Moré profiles test whether the Gurobi speedups persist across the heterogeneous instance pool rather than only in averages.



(a) Total-time speedup distribution.



(b) Performance profile.

Figure S4: At 0.1% MIPGap, the CPLEX visual evidence remains mixed under the stricter tolerance, supporting the solver-dependent interpretation of the main text.

Table S1: Hyperparameter sensitivity of SA-Aug (Simple class, Gurobi, 1% MIPGap). Solver-time sgm (s = 1 s) and PAR10, in seconds; default settings in bold.

Parameter	Value	sgm (s)	PAR10 (s)
Band u	1.5	2.09	2.46
	2	2.07	2.45
	4	2.12	2.49
Window $[\varepsilon_{\min}, \varepsilon_{\max}]$	$[10^{-6}, 10^{6}]$	2.04	2.40
	$[\mathbf{10^{-9}}, \mathbf{10^{9}}]$	2.02	2.39
	$[10^{-12}, 10^{12}]$	2.01	2.36

Base sgm is 3.73 s for reference. Each row is a separate matched run; the spread across all settings (sgm $\in [2.01, 2.12]$) is far below the base-to-scaled gap, and the median paired speedup is $\approx$ 1.0 throughout. The number of Ruiz iterations K is a parameter of the Ruiz baselines, not of the structure-aware rule, so it is fixed at $K = 4$; varying it leaves SA-Aug unchanged, as expected.

Table S2: Per-variant original-scale reliability breakdown of the four operational campaigns (the main text's original-scale audit table pools the scaled variants). For each solver, MIPGap, class, and variant: number of completed runs failing the strict absolute criterion ($v_{\text{row}} > 10^{-4}$ or $v_{\text{int}} > 10^{-5}$; n_{fail}) and the mixed absolute–relative criterion (additionally requiring the residual to exceed 10^{-6} relative to the violated row's magnitude; $n_{\text{fail}}^{\text{mix}}$); median and maximum row violation over completed runs; and PAR10 under the three readings (intent-to-treat as returned; strict, failures penalized as timeouts; mixed). Timeouts: 600 s for Simple, 3600 s otherwise; base-column PAR10 values are in the main-text tables.

Solver	Gap	Class	Variant	n_{fail}	$n_{\text{fail}}^{\text{mix}}$	v_{med}	v_{max}	PAR10 (ITT / strict / mixed)		
Gurobi	1%	Simple	SA-Aug	12	12	1.8×10^{-6}	3.7×10^{-3}	3	368	368
Gurobi	1%	Simple	SA-Mat	19	18	1.4×10^{-6}	1.9×10^{-3}	2	561	532
Gurobi	1%	Simple	Ruiz	55	52	1.2×10^{-5}	4.9×10^{-3}	27	1689	1599
Gurobi	1%	Simple	Ruiz+Cols	19	10	5.9×10^{-6}	2.8×10^{-3}	55	625	355
Gurobi	1%	SG-Ter-Mer	SA-Aug	1	0	6.1×10^{-7}	1.6×10^{-4}	562	1100	562
Gurobi	1%	SG-Ter-Mer	SA-Mat	1	0	5.9×10^{-7}	5.9×10^{-4}	21	558	21
Gurobi	1%	SG-Ter-Mer	Ruiz	1	0	6.9×10^{-7}	5.9×10^{-4}	567	1095	567
Gurobi	1%	SG-Ter-Mer	Ruiz+Cols	1	0	1.9×10^{-6}	1.1×10^{-1}	585	1113	585
Gurobi	1%	Full	SA-Aug	6	6	5.7×10^{-6}	2.6×10^{-3}	3308	9640	9640
Gurobi	1%	Full	SA-Mat	2	2	2.9×10^{-6}	2.8×10^{-4}	2487	4665	4665
Gurobi	1%	Full	Ruiz	2	2	3.1×10^{-6}	1.7×10^{-4}	4436	6522	6522
Gurobi	1%	Full	Ruiz+Cols	1	0	4.8×10^{-6}	8.5×10^{-4}	4696	5777	4696
Gurobi	0.1%	Simple	SA-Aug	21	17	1.9×10^{-6}	2.3×10^{-3}	109	752	631
Gurobi	0.1%	Simple	SA-Mat	30	26	1.5×10^{-6}	3.0×10^{-3}	76	954	837
Gurobi	0.1%	Simple	Ruiz	44	40	6.5×10^{-6}	4.9×10^{-3}	68	1391	1272
Gurobi	0.1%	Simple	Ruiz+Cols	20	10	6.8×10^{-6}	1.2×10^{0}	131	742	437
Gurobi	0.1%	SG-Ter-Mer	SA-Aug	1	0	6.2×10^{-7}	1.6×10^{-4}	594	1165	594
Gurobi	0.1%	SG-Ter-Mer	SA-Mat	1	0	5.8×10^{-7}	1.0×10^{-3}	22	559	22
Gurobi	0.1%	SG-Ter-Mer	Ruiz	0	0	6.7×10^{-7}	3.8×10^{-6}	587	587	587
Gurobi	0.1%	SG-Ter-Mer	Ruiz+Cols	1	0	1.9×10^{-6}	1.1×10^{-1}	629	1209	629
Gurobi	0.1%	Full	SA-Aug	2	2	5.0×10^{-6}	2.8×10^{-4}	3890	6283	6283
Gurobi	0.1%	Full	SA-Mat	3	1	2.0×10^{-6}	2.4×10^{-4}	4694	7769	5750
Gurobi	0.1%	Full	Ruiz	2	2	3.9×10^{-6}	3.4×10^{-4}	4976	7257	7257
Gurobi	0.1%	Full	Ruiz+Cols	2	1	6.8×10^{-6}	2.1×10^{-4}	6140	8451	7294
CPLEX	1%	Simple	SA-Aug	167	167	3.2×10^{-4}	4.4×10^{-1}	3	4912	4912
CPLEX	1%	Simple	SA-Mat	161	161	2.7×10^{-4}	4.4×10^{-1}	3	4736	4736
CPLEX	1%	Simple	Ruiz	160	160	2.9×10^{-4}	3.0×10^{-1}	4	4707	4707
CPLEX	1%	Simple	Ruiz+Cols	67	66	1.6×10^{-5}	1.1×10^{-3}	3	1972	1943
CPLEX	1%	SG-Ter-Mer	SA-Aug	0	0	6.6×10^{-7}	4.7×10^{-6}	10	10	10
CPLEX	1%	SG-Ter-Mer	SA-Mat	0	0	2.4×10^{-7}	1.4×10^{-6}	6	6	6
CPLEX	1%	SG-Ter-Mer	Ruiz	0	0	6.9×10^{-7}	5.4×10^{-6}	10	10	10
CPLEX	1%	SG-Ter-Mer	Ruiz+Cols	0	0	1.9×10^{-6}	1.4×10^{-5}	15	15	15
CPLEX	1%	Full	SA-Aug	9	9	2.2×10^{-5}	8.4×10^{-4}	5546	15014	15014
CPLEX	1%	Full	SA-Mat	8	7	1.4×10^{-5}	1.2×10^{-3}	6498	14932	13875
CPLEX	1%	Full	Ruiz	5	5	2.6×10^{-5}	4.9×10^{-4}	3506	8744	8744
CPLEX	1%	Full	Ruiz+Cols	2	1	7.8×10^{-6}	3.7×10^{-4}	4681	6793	5737
CPLEX	0.1%	Simple	SA-Aug	164	164	3.4×10^{-4}	7.9×10^{2}	41	4855	4855
CPLEX	0.1%	Simple	SA-Mat	165	165	2.8×10^{-4}	7.9×10^{2}	72	4915	4915
CPLEX	0.1%	Simple	Ruiz	159	159	2.9×10^{-4}	3.0×10^{-1}	71	4737	4737
CPLEX	0.1%	Simple	Ruiz+Cols	59	58	1.5×10^{-5}	3.9×10^{2}	105	1838	1808
CPLEX	0.1%	SG-Ter-Mer	SA-Aug	0	0	6.7×10^{-7}	5.6×10^{-6}	21	21	21
CPLEX	0.1%	SG-Ter-Mer	SA-Mat	0	0	2.4×10^{-7}	2.5×10^{-6}	16	16	16
CPLEX	0.1%	SG-Ter-Mer	Ruiz	0	0	7.0×10^{-7}	3.8×10^{-6}	8	8	8
CPLEX	0.1%	SG-Ter-Mer	Ruiz+Cols	0	0	2.0×10^{-6}	1.4×10^{-5}	539	539	539
CPLEX	0.1%	Full	SA-Aug	9	9	2.5×10^{-5}	2.4×10^{-4}	8930	18283	18283
CPLEX	0.1%	Full	SA-Mat	8	7	1.8×10^{-5}	3.2×10^{-4}	11186	19803	18718
CPLEX	0.1%	Full	Ruiz	2	2	1.7×10^{-5}	5.9×10^{-4}	5857	7917	7917
CPLEX	0.1%	Full	Ruiz+Cols	2	1	9.9×10^{-6}	5.8×10^{-4}	11222	13398	12310

Table S3: Out-of-sample sensitivity of the reliability floor for the collective coupling factor (SA-Aug), on twenty seeds (21–40) never used while designing the floor. Cells: coupling, coupling-uniform, and mixed patterns at severity $S = 6$ (60 instances, Gurobi, MIPGap 10^{-3}, 600 s limit). Median post-scaling condition number per pattern; number of runs failing original-scale verification; worst original-scale row violation; instances whose objective disagrees with the base beyond the MIPGap. The failure mode replicates out-of-sample with the floor disabled, and every floor value across three orders of magnitude eliminates it.

Floor	coupling	coupling-unif.	mixed	n_{fail}	$v_{\max}$	over gap
(pre-scaling)	4.4×10^8	3.3×10^6	1.5×10^{14}			
disabled	3.8×10^7	5.4×10^3	2.9×10^7	4	4.2×10^{-1}	4
10^{-2}	9.1×10^5	5.4×10^3	2.9×10^6	0	1.2×10^{-7}	0
10^{-3} (default)	3.5×10^5	5.4×10^3	2.3×10^6	0	1.2×10^{-7}	0
10^{-4}	1.0×10^6	5.4×10^3	2.3×10^6	0	1.2×10^{-7}	0
10^{-5}	2.1×10^6	5.4×10^3	3.0×10^6	0	1.2×10^{-7}	0

Table S4: Optional local reliability floor for SA-Aug, evaluated on the twelve Simple instances (Gurobi, 1% MIPGap) whose SA-Aug runs fail original-scale verification. The floor requires the applied local factor $d_r \geq \varepsilon_{\text{floor}}$, bounding the original-scale residual by $\varepsilon_{\text{solver}}/\varepsilon_{\text{floor}}$; rows below the floor stay at base scale. n_{fail} is the number of the twelve runs failing the strict absolute criterion; σ_{gm} is the shifted geometric mean solver time ($s = 1$ s); κ^{post} is the median exported condition number; $v_{\max}$ is the worst original-scale row violation. SA-Mat is shown for reference.

Variant	n_{fail}	σ_{gm} (s)	κ^{post}	$v_{\max}$
Base	0	4.26	–	0
SA-Aug (no floor, default)	12	2.29	6.6×10^9	3.7×10^{-3}
SA-Aug + local floor 10^{-3}	6	4.64	2.0×10^{14}	4.0×10^{-2}
SA-Aug + local floor 10^{-2}	0	2.63	5.5×10^{11}	7.4×10^{-5}
SA-Mat	5	2.57	4.5×10^{12}	1.9×10^{-3}

Notes. On this failing subset the local floor at 10^{-2} eliminates every verification failure while retaining a solver-time advantage over the base (σ_{gm} 2.63 vs 4.26 s), even though it *raises* the exported condition number (the protected rows are exactly the over-compressed large-right-hand-side rows)—a further illustration that exported conditioning is a weak predictor of runtime. The floor at 10^{-3} is too shallow (six failures remain and the residual bound is only $\varepsilon_{\text{solver}}/10^{-3}$). The floor is off by default; a full-class runtime characterization across all Simple and Full instances is the natural next step.